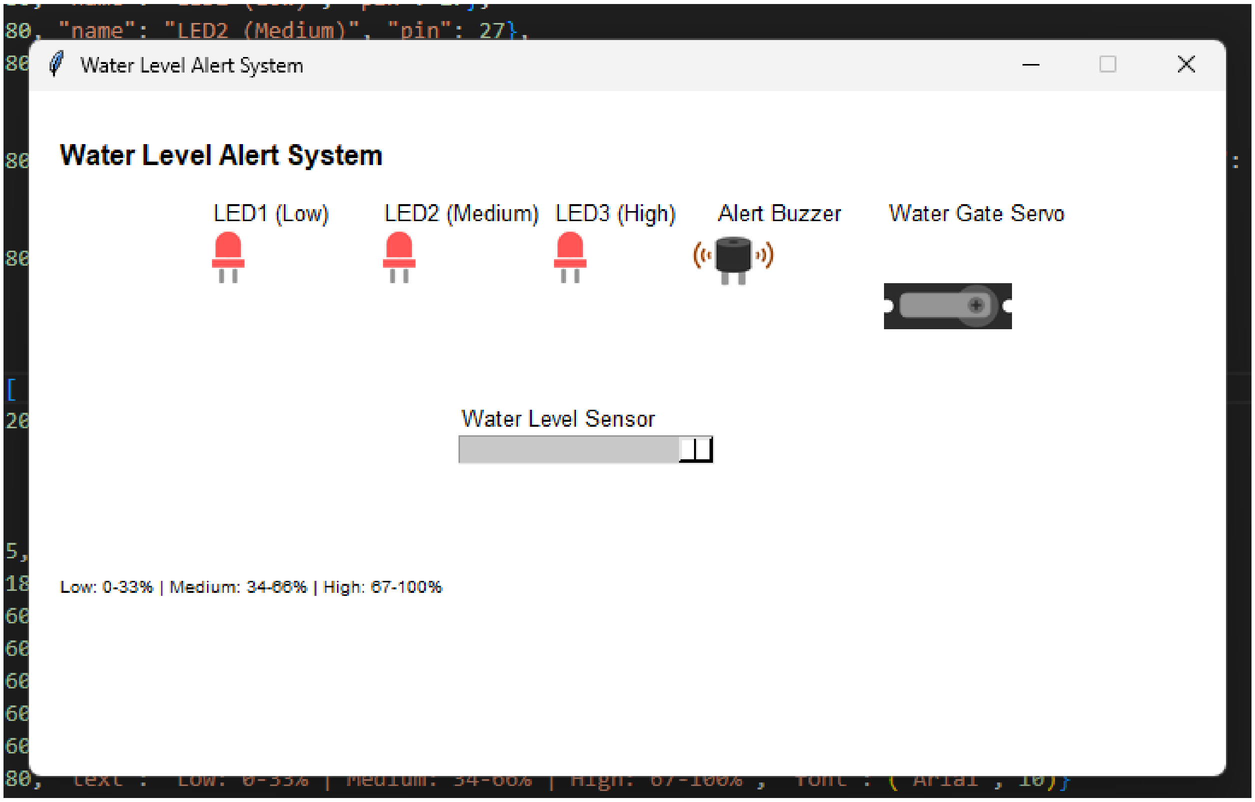
6614022005 ธนวัฒน์ เหลืองวิโรจน์



water_alert.py

```
from gpiozero import Servo, MCP3008, Button, LED, Buzzer
from time import sleep
import signal

print("Water Level Alert System is running...")

led_low = LED(17)
led_medium = LED(27)
led_high = LED(22)
servo = Servo(24)
buzzer = Buzzer(25)
water_sensor = MCP3008(0)

def main():
  print("System active. Press Ctrl+C to exit.")

  while True:
    water_level = water_sensor.value
    print(f"Water level: {water_level:.2f} ({int(water_level * 100)}%)")

    # (Low) (0-33%)
    if water_level >= 0:
      led_low.on()
    else:
      led_low.off()

    # Medium (34-66%)
    if water_level > 0.33:
      led_medium.on()
```

```
    else:
      led_medium.off()

    # High (67-100%)
    if water_level > 0.66:
      led_high.on()
      buzzer.on()
    else:
      led_high.off()
      buzzer.off()

    servo_position = (2 * water_level) - 1
    servo.value = servo_position

    sleep(0.5)

if __name__ == "__main__":
  main()
```

water_alert_simulation.py

```
from tkgpio import TkCircuit

configuration = {
  "name": "Water Level Alert System",
  "width": 700,
  "height": 400,

  "leds": [
    {"x": 105, "y": 80, "name": "LED1 (Low)", "pin": 17},
    {"x": 205, "y": 80, "name": "LED2 (Medium)", "pin": 27},
    {"x": 305, "y": 80, "name": "LED3 (High)", "pin": 22}
  ],
  "servos": [
    {"x": 500, "y": 80, "name": "Water Gate Servo", "pin": 24, "min_angle": -90, "max_angle": 90,
"initial_angle": 0}
  ],
  "buzzers": [
    {"x": 400, "y": 80, "name": "Alert Buzzer", "pin": 25}
  ],
  "adc": {
   "mcp_chip": 3008,
   "potentiometers": [
    {"x": 250, "y": 200, "name": "Water Level Sensor", "channel": 0}
   ]
  },
  "labels": [
    {"x": 15, "y": 25, "text": "Water Level Alert System", "font": ("Arial", 16, "bold")},
    {"x": 250, "y": 180, "text": "Water Level Sensor", "font": ("Arial", 10)},
    {"x": 105, "y": 60, "text": "Low", "font": ("Arial", 10)},
    {"x": 205, "y": 60, "text": "Medium", "font": ("Arial", 10)},
```

```
        {"x": 305, "y": 60, "text": "High", "font": ("Arial", 10)},
        {"x": 500, "y": 60, "text": "Water Gate", "font": ("Arial", 10)},
        {"x": 400, "y": 60, "text": "Buzzer", "font": ("Arial", 10)},
        {"x": 15, "y": 280, "text": "Low: 0-33% | Medium: 34-66% | High: 67-100%", "font": ("Arial", 10)}
    ]
}

circuit = TkCircuit(configuration)

@circuit.run
def main():
    import water_alert

    water_alert.main()
```